Research Assigment 1: Data Analytics & Data Science Foundations

1. A database is an organized collection of information that is stored and managed in a structured way so it can be easily accessed, updated and processed.
In simplier terms, it is a system that helps us store retrieve and manage data efficiently. ~~without~~

Databases are essential in the modern world because almost every digital system depend on data.
They help organizations streamline information, improve efficiency, ensure accuracy and support decision making
Without databases, it would be extremely difficult to manage large amounts of information such as customer records, financial transaction, or medical data.

1. Banking System => Banks use databases to store customer account details, track transactions and manage balances securely and accurately.
2. Social Media Platforms => Apps like instagram or Facebook use database to store user profile, posts messages, and interactions in real time.

2. A data warehouse is a large, centralized system that stores data collected from multiple sources for the purpose of analysis and reporting.
Unlike a regular storage that focuses on daily operations, a data warehouse is designed to support

long-term decision making and strategic analysis.

The main difference between a data warehouse and regular database is their purpose.
A regular database is used for day-to-day operation such as processing transactions, updating records and handling real-time data.
In contrast a data warehouse stores historical and consolidated data, which is structured specifically for analysis rather than routine operations

A company would use a data warehouse because it allows them to combine data from different systems analyze trends, and make informed business decision
In simple terms, a data warehouse turns raw data into meaningful insights that supports smarter decision making.

3. A data lake is a centralized repository that stores vast amounts of raw data in its original format

The key difference Key difference between a data lake and a data warehouse lies in how the data is stored and used.
A data warehouse stores cleaned, structured and organized data that is ready for analysis, making it ideal for reporting and business intelligence.
A data lake however store raw and unprocessed data, which can be transformed and analyzed only when needed.
In simple terms, a data

prepared data while a data lake focuses on
and large-scale data storage

A business would prefer a data lake when dealing with large volumes of diverse data, especially when the future use of that data is not yet clearly defined.
It is particularly useful for advanced analytics, machine learning, and big data projects, where data scientist may need access to detailed and unfiltered information.
By using data lake, companies are able to retain all their data, support experimentation and extract deeper insights over time

4. A data mart is a smaller, specialized subset of data that is designed to serve a specific department or business function within an organization.
It contains focused and relevant data, making it easier for users to access and analyze information quickly and efficiently

A data mart is closely related to a data warehouse because it is often ~~dene~~ derived from it.
While a data warehouse stores large volumes of centralized and comprehensive data for the entire organization, a data mart extracts only the relevant portion of that data for a particular team, such as marketing, finance or sales.
In this way, a data mart simplifies access by providing a more targeted and streamline view of the data.

Data marts are typically used by business professionals and departmental teams, such as analysts, manager and decision makers who needs specific insights without dealing with the complexity of the full data warehouse. This allows them to analyze data and more efficiently improve performance, and make informed decisions within their specific area of responsibility

Data can be categorized into structured, semi-structured and unstructured data based on how it is organized and stored.
Understanding these differences is essential because each type requires different methods for storage, processing and analysis

Structured data refer to data that is highly organized and stored in a fixed format, typically in rows and columns. It is easy to store, search and analyze using traditional database
Example: A spreadsheet or database table containing customer names, contact details and account balance

Semi-structured data is not strickly organized in rows and columns, but it still contains some level of structure through tags, labels or markers.
This makes it more flexible than structure data while still being partially organized
Example: JSON or XML files used in web applications where data is stored with tags like name, age and address.

Unstructured data refers to data that has no predefined format or organization, making it more complex to process and analyze.
It often includes large amounts of text or media
Example: Social media posts, images, video or audio recordings.

In summary, structured data is organized and easy to analyze, semi-structured data is partially organized and flexible and unstructured data is raw and complex requiring more advanced tools for analysis

Data Modelling is the process of defining and structuring how data is organized, stored, and related within a system.
It provides a clear blueprint of how data elements interact, ensuring that information is logically arranged and efficiently managed.

Data Modelling is important because it helps designers create accurate, efficient and scalable systems.
By establishing clear relationships and structures, it minimizes errors, reduces redundancy and improves data consistency.
It also enhances communication between developers analysts and stakeholders by providing a shared understanding of how the system will handle data.
Ultimately, effective data modelling ensures that a system can store, retrieve and process data reliably while supporting informed decision-making

Types of data models:

⇒ Conceptual Data Model

This is a high level ~~rep~~ representation of data that focus on the overall structure rather than technical details

It defines key entities and their relationship, making it useful for understanding the system from a business perspective

Logical Data Model

This model provides more detail by defining attributes data types and relationships without considering how the data will be physically stored.

It acts as a bridge between the conceptual design and the physical implementation

7. Data mining is the process of analyzing large volumes of data to discover patterns, trends, and meaningful insight that are not immediately obvious.

It involves using statical methods, algorithms and machine learning techniques, to transform raw data into valuable information that can support decision making.

In business data mining is used to enhance performance, optimize strategies and predict future outcomes By identifying hidden patterns in data, organizations can better understand customer behavior, improve operations and gain a competitive advantag.

It enables companies to make data-driven decisions rather than relying on assumptions.

For Example

- Retail & E-commerce
Businesses analyze customer purchase history to identify buying patterns and preference.
This allows them to recommend products, and personalize marketing campaigns and increase sales
Example ; An online store suggest products based on what customer previously viewed or purchased.

- Banking and Fraud Detection
Financial institutions use data mining to detect unusual transaction patterns that may indicate fraud.
By analyzing spending behaviour, they can identify suspicious activities and prevent financial losses
Example ; A bank flagging a transaction made in a different country shortly after a local purchase

In summary, data mining help business extract valuable insight from large datasets, enabling smarter decision, improve efficiency and better customer experience

8. A Database Management System (DBMS) is a software that is used to create, manage store and control access to databases.
It acts as the interface between users and the database, allowing data to be efficiently organized, updated, retrieved and secured.
A DBMS is what makes it possible to work with large amounts of data in a structured and reliable way

(a) My SQL
Is an open-source relational DBMS that uses structured tables and SQL (Structured Query Language)
It is widely used because it is reliable fast and easy to use
Small to medium-sized web application such as blogs e-commerce website and content management system.

(b) PostgreSQL
This is an advanced open-source relational DBMS known for its strong support for complex queries and data integrity. It is highly extensible and supports advanced data types
Applications that require complex data analysis, such as financial systems, geospatial applications and analytics platforms.

) Oracle Database
Oracle is a powerful, enterprise-level DBMS designed for large organizations.
It is known for its high performance of critical data
Large corporations like banks, insurance companies and government systems that handle massive volumes of critical data.

Mongo DB
Is a NoSQL DBMS that stores in flexible, JSON-like documents instead of tables. It is designed for handling structured or semi-structured Data
real-time applications such a mobile apps, content management systems and applications that require flexible scalable data storage

ETL (Extract, Transform, Load)
Is a process used data analytics to collect data from different sources, convert it into a usable format and store it in a central system such as a data warehuse.
It is a critical process that ensures data is accurate, consistent and ready for analysis.

The three steps of ETL
Extract
step 1: This step involves collecting or retrieving data from various sources, such as databases, website or applications
This data may come in different formats and structures.
Example: Extracting sales data from an online store and customer data from a CRM system.

Step2: Transform ⇒ In this stage, the extracted data is cleared organised and converted into a consistent format.
This may include removing errors, correcting inconsistencies and combining data from multiple sources.
Example: Standardizing date formats, removing duplicates records and categorizing customer information

Step3 Load: The step involves storing the transformed data into a target system, such as a data warehouse where it can be accessed for analysis and reporting

ETL is essential because it ensures that data is reliable accurate and ready for decision-making.
It allows organizations to integrate data from multiple sources, improve data quality and support meaningful analysis.

Without ETL, business would struggle with inconsistent and unorganized data, making it difficult to generate insight or make informed decisions

ETL helps transform raw data into valuable, structured information, enabling businesses to analyze trends, improve performance and make data driven decisions

CSV (Comma - Separated Values)
Is a simple, text-based format where data is stored in rows and columns, separated by commas. It is easy to read and widely supported across tools
When to use: When working with structured data, especially for small to medium datasets, data sharing or simple analysis task

JSON (Java Script Object Notation)
Is a semi-structure format that stores data in key-value pairs. It is flexible and commonly used in web applications.
When to use : When handling semi-structured data, APIs or applications that require flexible and human readable data formats.

Parquet
Parquet is a columnar storage format optimized for big data processing. It is highly efficient in terms of storage and performance, especially for analytical queries
When to use: When dealing with streaming data or system that require frequent schema changes, such as real-time data pipelines.

Avro
Avro is a row-based data format that supports schema evolution, meaning the data structure can change over time without breaking the system. It is compact and efficient for data serialization.
When to use: When dealing with streaming data or system that require frequent schema changes, such as real-time data pipeline

11. Artificial Intelligence (AI)
Refers to the ability of computer systems to simulate human intelligence in order to perform tasks such as learning reasoning, problem-solving and decision making.
AI systems are designed to analyze data, recognize patterns and generate insights, thereby enhancing efficiency and supporting informed decisions across various industries

The distinction between Narrow AI (Weak AI) and General AI (Strong AI) lies in their scope and capabilities

Narrow AI (Weak AI) is designed to perform a specific task or a limited range of tasks.
It operates within predefined boundaries and cannot function beyond its programmed purpose
Despite its limitations, it is highly effective and widely utilized in real-world applications
Virtual assistance such as Siri or recommendation system used by streaming platforms to suggest content based on user preference.

General AI (Strong AI) in contrast, refers to a more

advanced form of intelligence that can understand learn and apply knoledge across a wide range of tasks, similar to human cognitive abilities.
It is capable of reasoning, adapting and solving unfamiliar problems without being explicitly programmed for each task
Example: A AI system that can perform any intellectual task a human can, such as learning a new language, solving complex scientific problems and making idependent decisions

12 Machine Learning (ML) is a subset of artificial intelligence that enables computer system to learn from data, identify patterns and make predictions or decisions without being explicitly programmed.
Instead of following fixed instructions, ML systems analyze large data sets, adapt over time and improve their performance automatically

The key difference between machine learning and traditional programming lies in how the system developed. In traditional programming developers define explicit rules and logic that the computer must follow to produce an output. In contrast, machine learning systems are provided with data and algorithms, allowing them to learn patterns and generate outputs idependently, thereby enhancing flexibility and scalability

Types of Machine Learning
Supervised Learning
This involves training a model on labelled data, where the input and the correct output are already known.
The system learns to map inputs to outputs and make accurate predictions on new data.

Example of Supervised Learning
Predicting house prices based on features such as size location and number of rooms

Unsupervised Learning
This works with unlabeled data, meaning the system must identify patterns, relationships or groupings without predefined outcomes
Example: Customer segmentation, where a business groups customs based on purchasing behaviour

Reinforcement Learning
This involves training a model through trial and error, where it recieves rewards or ~~penalties~~ penalties based on its actions. The system learns to optimize decisions over time to achieve the best outcome.
Example: Training a robot or AI agent to navigate an environment to play a game by maximizing rewards

Machine Learning differs from traditional programming because it learns from data rather than relying solely on predefined rules, enabling systems to become more accurate, adaptive and efficient over time.

Deep Learning is a specialized subset of machine learning that utilizes neutral network with multiple layers to analyze large volumes of data and identify intricate patterns.
It is designed to process unstructured data such as images, audio and text thereby enabling more advanced and accurate predictions

Deep Learning exists within the broader of artificial intelligence. Artificial Intelligence (AI) represents the overall concept of machine simulating human intelligence, while Machine Learning is a subset that allows systems to learn from data.
Deep Learning further refines this approach by using layered neural networks to enhance learning capability and performance.
In simple terms, deep learning is a more advanced technique that optimizes how machines learn from complex data.

Neural Networks are the fundational technology behind deep learning. They computational models inspired by the structure of the human brain, consisting of interconnected nodes (neurons) organized into layers: an input layer one or more hidden layer and an output layer

Neural networks functions by receiving input data, processing it through multiple layers and generating an output.
Each connection between neurons has a weight that determines its importance. As data passes through the network, these weights are adjusted through a learning process, allowing the system to improve accuracy and refined predictions over time.
This process enables neural networks to recognize patterns classify data and make informed decisions

4. Natural Language Processing (NLP)
Is a branch of artificial intelligence that enables computers to understand, interpret and generate human

language in a meaningful way. It allows machines to process text and speech data, thereby bridging the gap between human communication and computer systems. NLP combines computational linguistics with machine learning to analyze language structure, extra meaning and generate appropriate response

Real-world application of NLP

(a) Virtual Assistants
Applications such as Siri, Google Assistant use NLP to interpret spoken commands, understand user intent and respond appropriately. They can perform tasks such as setting reminders, answering questions and controlling smart device.

(b) Search Engine
Search Engines like Google use NLP to understand search queries and deliver relevant relevant results, even when the input is not perfectly structured. This allows users to find acurate information quickly and efficiently

(c) Translation Tools
Tools such as Google Translate use NLP to convert text or speech from one language to another, enabling communication across different languages in real time.

NLP enhance communication between humans and machines by enabling systems to process, understand and respond to natural language, making everday everyday digital interactions more seamless and intelligen

A Large Language Model (LLM) is an advanced type of artificial intelligence model that is trained on massive amounts of text data in order to understand, generate and manipulate human language.
It uses deep learning techniques, specifically transformer based neural networks, to predict and produce coherent context-aware text based user input.

The key difference between a Large Language Model and standard machine learning model lies in their scope complexity and capabilities.
A standard machine learning model is typically designed to perform a specific, narrow task such as predicting house prices or classifying emails.
In contrast LLM is trained on extremely large and diverse datasets, enabling it to perform a wide range of language related tasks such as answering questions, writing essay summarizing information and generating code.

Examples of well know LLMs:
ChatGPT (OpenAI models)
ChatGPT is a large language model used to generate human like text, answer questions, assist with writing and support to improve productivity and communication
Google Gemini
Is an advanced LLM developed by Google that is used to understand ~~ang~~ and generate text, assist with research summarize information and support complex reasoning tasks across different formats such as text and images.

Large Langtag Models are powerful AI systems that specialize in understanding and generating human language at scale, making them significantly more versatile than standard machine learning models, which are usually limited to a specific tasks.

Generative AI is a branch of artificial intelligence that ~~forces~~ focuses on creating new content, such as text images, audio, video and code based patterns learned from datasets.
Instead of only analyzing and classifying data generative AI systems are designed to produce original outputs that resemble human created content.

At a high level generative AI works by using deep learning models trained on massive amounts of data. During training, the model learns patterns, structures and relationships within data.
When a user provides a promt, the model predicts and generates the most likely sequence of word, pixels or sounds that match the request.
This process allows the system to produce contextually relevant and realistic outputs.

Three tools that use Generative AI
a) Chat GBT (OpenAI)
Is a generative AI tool that produce human-like text response. It can write essay, answer questions, summarize information and assist with problem-solving by generating context-aware written content

Github Copilot is a generative AI tool designed for programming. It generate code suggestions and complete functions automatically based on the developer's input, helping to speed up software development.

Generative AI works by learning from large datasets and using them to create new, original content, making it a powerful tool for writing, designing, coding and creative problem-solving

11. A training dataset is a collection of data used to teach a machine learning or AI model how to perform a specific task. It contains examples that the model learns from, allowing it to identify patterns, relationships and trends so that it can make accurate predictions or decisions on new, unseen data.

The quality of training data is extremely important because an AI model is only as good as the data it learns from. High quality data ensures that the model can learn accurate patterns, generalize effectively and produce reliable outputs.
If the data is clean, balanced and representative of real-world scenarios, the model will perform more efficiently and consistently.
In contrast poor quality data leads to inaccurate learning and weak performance

If the training data is biased or incorrect, several problems can occur. The model may learn and reinforce unfair patterns, leading to bias or discriminatory outcomes. It may also produce inaccurate or misleading predictions

reducing trust in the system. In some case, the model m
completely fail to perform correctly in real-world situations
because it has learned from flawed information. This can have
serious consequences, especially in sensitive areas such as
healthcare, finance and hiring systems.

18 Computer vision is a field of artificial intelligence that
enable machine to interpret, analyze and understand visual
information from the world, such as images and video. In simp
terms, it allow computers to "see and make sense of visual
data in a way that is similar to human vision

Machines do not see images the way we do. Instead the
process ~~data~~ images as mathematical ~~data~~ made up of
pixels. Computer vision systems use algorithms and
deep learning models to detect patterns, shaps, edges, colors
and textures within these pixels. Through training on large
datasets of labeled images, the system learns to recognize
objects, classify scenes and identify features with increa-
sing accuracy

Real-world use cases of Computer Vision:
Facial Recognition System
Computer vision is used in security system and smart-
phones to identify or verify a person's identity based
on facial features. It analyzes facial patterns and
matches them with stored data for authentication
purposes.

elf-Driving Cars
utonomous vehicles use computer vision to detect road
lanes and obstacles. This allows ~~the~~

the car to make real-time driving decisions and navigate safely without human intervention

Computer vision enables machines to extract meaningful information from visual data, transforming images and videos into actionable insights used in security, transportation, health and many other industries

19 An API (Application Programming Interface) is a set of rules and protocols that allows different software systems to communicate, exchange data and work together. In simple terms, an API acts as a bridge or messager between two applications, enabling them to interact without needing to know how each other is built internally.

Simple real-world analogy
An API can be compared to a waiter in a restaurant
⇒ You (the user) place an order (request) with the waiter (API)
⇒ The waiter takes your request to the kitchen (system /database)
⇒ The kitchen prepares the food (process the data)
⇒ The waiter brings the food back to you (response)

You do not need to know how the kitchen operates; you only interact with the waiter. Similarly, applications use APIs to request and recieve information without accessing the internal system directly.

APIs are essential because they enable integration, efficiency and scalability in software and data systems

They allow different applications, platforms and services to work together seamlessly, even if they were built using different technologies. This makes it possible for systems to share data in real time, automate process and extend functionality

APIs allows: Mobile apps to connect to cloud database
Websites to display live weather or map data
Payment systems to process transactions securely

20. An API key is a unique identifier (usually a long string of letters and numbers) that is used to authenticate and authorize access when a user or application connects to an API. In simple terms, it acts as a digital security password that confirms who is making a request and whether they are allowed to access the system.

The main purpose of an API key is to control and secure access to an API. It helps the system:

- Identify the application or user making the request
- Monitor and track API usage
- Restrict access based on permission or usage limits
- Prevent unauthorized access to sensitive data or service

How API key should be stored and protected
API keys must be treated as highly sensitive credentials. They must be:

- Stored in secure environments such as environment variables or secure vaults
- Kept out of source code and public repositories (like GitHub)
- Restricted with proper access controls and permissions
- Regular rotated or updated for security purposes

What happens if an API key is exposed
If a API key is exposed publicly, it can lead to serious risks such as:

- Unauthorized access to system or data
- Abuse of services, leading to unexpected cost or resources exhausion
- Data breaches, where sensitive information may be accessed or ~~stolt~~ stolen
- Service disruption, if attacks overload the API with ~~acc~~ excessive request

An API key is a critical security credential used to control access to APIs and it must be carefully protected. If exposed, it can compromise system security, financial resources and data integrity

21 A Rest API (Representational State Transfer Application Promming Interface) is a type of web service that allows different system to communicate over the internet using standard HTTP methods.
It is designed around a set of principle that make systems simple, scalable and easy to integrate, especially in web and mobile applications.

Rest stands for Representational State Transfer. It refers to a software architecture style where resources (such as data or services) are accessed and manipulated using standard web protocols, typically through URLs

Main HTTP methods used in REST APIs
GET
The GET method is used to retrieve or read data from

a server without modifying it.
Example : Fetching a list of users or product information from database

POST
The POST method is used to send data to the server to create a new resource
Example : Submitting a registration form to create to a new user account

PUT
The PUT method is used to update or replace an existing resource on the server
Example : Updating a user's profile information, such as email or password

DELETE
The DELETE method is used to remove a resource from the server
Example : Deleting a user's profile information, such as email or password.
Deleting a user account or removing a product from database

REST API enables communication between systems using standard HTTP methods, where each method has a specific role in retrieving, creating, updating or deleting data, making it a fundamental part of modern web development and data systems.

JSON (Javascript Object Notation) is a lightweight based format used to store, structured, and exchange data

Promt Engineering is the practice of designing and structuring input instructions (prompts) in a clear and effective way to guide AI system in producing accurate, relevant and high-quality.
It involves carefully choosing wording, context and constraints so that the AI can interpret the request correctly and generate the desired responds

Prompt engineering is an impertant skill when working with AI tools such as ChatGPT or Claude because these system rely heavily on the quality of the input provided.
A well-structured promt leads to more precise, useful and contextually appropriate response, while a vague or poorly written prompt can result in incomplete or irrelevant outputs
In essence, better prompt lead to more precise, useful and contextually appropriate response, while a vague or poorly written promt can result in incomplete or irrelevant outputs.

Poor Promt : "Write about AI"
Improve Promt: "Write a clear and structured explanation of Artificial Intelligence, including its definition, main types and two real-world applications using simple but professional language

Promt Poor : "Help me with data"
Promt Improved : "Explain what data analysis is and describe the main steps involved in analysing datasets including data collection

A map function is a higher-order function used in programming and data ~~functions~~ processing to transform

each element in a collection (such as a list or array) by applying a specific operations to every item. It then returns a new collection with the transformed values without modifying the original data.

In essence, the map function is used to apply a consistent transformation at scale, making it highly efficient for working with large datasets. It is commonly used in functional programming and data analytics to clean modify or standard data.

How it works

- Take an input list or array
- Applies a function to each element
- Returns a new list with updated values.

Python

```
numbers = [1, 2, 3, 4, 5,]

square_numbers = list (map (lambda x: x* x, numbers))
print (squad_numbers)
```

Output

[1, 4, 9, 16, 25]

25. Cloud Computing refers to the delivery of computing services such as storage, servers, databases and software over the internet rather than relying on local machines or physical infrastructure.
It enable organizations to access, manage and scale resources on demand, thereby enhancing flexibility efficiency and cost effectiveness

Three main cloud service models

IaaS (Infrastructure as a Service)

IaaS provides ~~vit~~ virtualized computing resources such as servers, storage and networking over the internet. Users have control over the infrastructure but are responsible for managing operating systems and applications

PaaS (Platform as a service)

PaaS provides a complete development and deployment environment, allowing developers to build, test and deploy applications without managing the underlying infrastructure. It simplifies development by handling system maintenance, scalability and runtime environments

Example: Google App Engine - used to develop web applications efficiently

SaaS (Software as a Service)

SaaS delivers fully function ~~soft~~ software applications over the internet, which users can access through web browser without installation or maintenace. The service provider manages everything including updates and security

Example Microsoft 365 - used for email document creation and collaboration tools.

86. A version ~~that~~ Control System (VCS) is a tool that enables users to track, manage and control change made to files over time. It allows multiple users to collaborate efficiently while maintaining a complete history of modifications. This ensures that changes can be reviewed reversed or compared, thereby enhancing accuracy and minimizing errors in projects.

Git is widely used distributed version control system that allows developers and data analyst to make code, collaborate with teams and maintain project history efficiently. It is popular because it is fast, reliable and supports both individuals and colaborative workflows. GitHub enables users to work on different features simultaneously, track changes precisely and intergrate updates, seamlessly making it essentials in modern software development and data projects.

Commit
A commit is a saved snapshot of changes made to a project at a specific point in time. It allows users to record progress, document updates and revert to previous versions if necessary. Each commit includes a message that describes the changes made

Branch
A branch is a seperate version of a project that allows users to develop features or test changes independently without affecting the main project. Once work is complete, the branch can be merged back into main version.

A Jupyter Notebook is an interactive web-based environment that allows users to write code, visualize data, and document their work in a single, intergrated interface. It combines executable code, text explanations and visual outputs, thereby enabling users to analyze data and present results in a clear and structured manner

Jupyter Notebook are most commonly used with the Python programming language, although they also support other language such as R and Julia. Python is predominatly used because of its powerful libraries for data analysis such as Pandas, Numpy and Matpotlib.

Why Jupyter Notebook are popular

a) Interactive and Flexible: They allow users to execute code step by step making it easier to test, debug and refine analyst

b) Intergrated documentation: Users can combine code with explanation, which helps to communicate insights and documents the analytical process efficiently

28 Python is a highly-level general-purpose programming language that is designed to be simple, readable and highly efficient. It is widely used in data analytics and artificial intelligence because it enable developers and analysts to write clear code, process data ~~efficiently~~ effectively and build powerful models with minimal complexity.

Python is considered one of the most popular language for data analytics and AI because it provides a rich ecosystem of libraries and tools that facilitate data processing, analysis, visualization and machine learning. In addition its syntax is intuitive, which allows users to develop solutions quickly enhance productivity and scale applications efficiently

manipulation and analysis
Pandas data structures such as DataFrames which allow users to organize, clean and analyze structured data efficiently

Numpy
Is a library designed for numerical computing. It enables fast operations on array and matrices thereby optimizing mathematical calculations and data processing performance

Matplotlib
Is a visualization used to create charts, graphs and plots. It helps user present data insights clearly through visual representation

Scikit-learn
Is a machine learning library that provides tools for building, training and evaluating models. It is widely used for tasks such as classification, regression and clustering

29. Data Privacy refers to the protection of personal and sensitive information, ensuring that it is collected such as ~~identity~~ identity theft or unauthorized access. It also enable organizations to build trust, maintain compliance with regulations and minimize legal and reputational risks.

The General Data Protection Regulation is a law that was introduced by the European Union to regulate how personal data is collected, processed and

protected. It aims to enhance data privacy right and ensure that organizations handle personal data transparently and securely.

GDPR applies to all countries within European Union Any organization worldwide that collects or processes personal data of individuals residing in the EU regardless of where the company is located.

30. Data Ethics refers to the principles and standard that guide how data is collected, used and managed responsible It ensures that practice are fair, transparent and respectful of individuals right, thereby helping organizations make decisions that are not only effective but also ethical

Example of unethical use of data or AI

(a) Biased AI in hiring systems
Some companies use AI to screen job applicants. However if the training data is biased, the system may favor $ certain groups while disadvantaging others
→ This is problematic because it can reinforce discrimination and limit equal opportunities even if the bias is ~~unintentional~~ tional

(b) Misuse of personal data Without consent
Organization may collect user data and use it for purposes that were clearly disclosed, such as selling it to third parties
This is ~~the~~ unethical because it ~~vio~~ violates user privacy and ~~the~~ trust and individuals lose control over how their personal information is used.

(e) Manipulative algorithms on social Media
AI systems can be designed to promote addictive or misleading content to keep users engaged for longer periods
→ This is problematic because it can influence behavior, spread misinformation and negatively impact mental well-bein prioritizing profit over user welfare

31. Data Governance refers to the overall framework of policies processes and standards that ensure data is managed, protected and used effectively within an organization.
It establishes clear guidelines on how data should be collected, stored, accessed and maintained, thereby ensuring consistency, security and reliability

Key components of a good data governance framework
(a) Data Quality Management
This ensures that data is accurate, complete and consistent High-quality data enables organization to make reliable and informed decision

(b) Data & security
This component focuses on protecting data from unauthorized access and breaches. It helps organizations safeguard sensitive information and comply with regulations, thereby minimizing risks

(c) Data Policies and Standards
These are formal rules that define how data should be handled. They ensure that data is used consistently across the organization, which enhances efficiency and reduces confusion.

Roles and Responsibilities
A strong framework clearly defines who is responsible for mana-
ging and maintaining data. This includes roles such as as
data owners and data steward, which help to ensure accoun-
tability and proper data management

Data Lifecycle Management
Data governance provides a structured approach to manage
and control data efficiently, ensuring that it is accurate
secure and used responsibly across organization

Biased in AI refers to a situation where an Artificial
Intelligence system produces systematically unfair or
skewed results, often favoring one group over another. An
AI model becomes biased when it learn from unbalanced
incomplete or prejudiced training data, or ~~the~~ when its
designed unintentionally reflects human assumptions

Real-world example of AI bias
Facial recognition systems have been shown to perform
less accurately on people with ~~darder~~ darker skin tone
compared to lighter skin tones. This happens because
the training datasets used to build these systems
contained more images of certain groups than other
leading to uneven performance.

This type of bias can lead to serious consequences such as:
- Incorrect identification of individuals
- Unfair targeting in security or surveillance system
- Discrimination in law enforcement to access control syst

23. Data Analyst
Interprets existing data to identify trends and supports business decision
Data cleaning, data visualization reporting, basic statistics communication
Excel, SQL, Power BI, Tableau, Python (Pandas)

Data Scientist
Build predictive models and extracts deeper insights from complex data
Machine learning statics, programming, data modelling experimentation
Python, R, Scikit-learn, TensorFlow, SQL

Data Engineer
Designs and ~~maintains~~ maintains system that collect, store and process large data sets
Database design, ETL pipelines cloud computing, big data processing
SQL, Apache, Spark, Hadoop, Python, AWS, Azure

AI Engineer
Develops and deploys AI systems such as machine learning and deep learning models
Deep learning, neural networks, model deployment, MLOps
Python, TensorFlow, PyTorch, OpenAI, APIs, cloud platforms

24. Business intelligence (BI) refers to the process of collecting analyzing and visualizing business data to help organizations make informed decisions, data-driven decisions. It transforms raw data into meaningful insights through reports, dashboards and charts, allowing business to understand performance and

identify trends

Two popular BI tools
Microsoft Power BI
Is a data visualization and reporting tool that allows users to create interactive dashboards and reports. It connect to multide data source and helps business track performance metrics in real time, making it easier to identify trends and make decisions quickly

Tableau.
Is a powerful BI tool used for creating visual ~~da~~ dashboards and analyzing complex datasets. It helps users explore data visually through charts, graphs and maps, enabling to understand patterns and insights without needing advanced technical skills

35. I found ~~AI~~ how AI ~~lear~~ connects with Machine Learning Deep Learning and Larg Language Models

What stood out is how AI systems learn patterns from large amounts of data and use those patterns to make predictions or generate outputs. It is surprising how something like a chatbot operates through mathematical patterns, probalities and training data.

36. The most difficults concept funny enough are the ones I now find most interesting which are AI, Machine Learning and Deep Learning it ~~Ne~~ was hard to crasp what each one does and how it does it especially ~~the~~ Deep learning understanding nueral networks and how data goes through different layers or how it process data in a more

complex way

37. Data and AI will likely have strong impact on both my future career and everyday life especially as I move into a field like data analytics and social media management

In my future career I can see AI and data helping me make faster and more informed decisions for example a ~~social media~~ ~~man~~ data analyst, I can use AI tools to analyze large datasets, identify patterns in customer behavior and predict future trends that can support business decisions. This means I will not rely on guesswork but instead use data-driven insight to improve accuracy in reporting and helping organizations make better strategic choices

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